ALGEBRAS OVER THE DERIVED OPERAD ARE SPECIAL

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Allowed standard prerequisites (editorial).

- The embedding of the derived operad into the free operad on the doubled variables. same article, the inclusion used at proof lines 744-746. Allows a presentation of an algebra over the derived operad to be transported into the larger free operad.
- The weight criterion for membership in the derived operad. same article, the lemma at source lines 562-568. Identifies the elements of the derived operad inside the free operad as the weight homogeneous ones, which is what makes the differential ideal generated by the relations coincide with the relations themselves.

Source reference. Original lemma internal: sources/sigma-2024-017-on-pre-novikov-algebras-and-derived-zinbiel/pinned-original-source.tex, source lines 562–568. The counted statement or proof refers to that the author's preceding lemma giving the weight criterion for the derived operad; it is the author's own earlier display, definition or statement in the same article and the whole argument of the counted span is retained verbatim.

Theorem 3.5. *If* Var *is a binary operad such that* $\mathrm{Var} \circ \mathrm{Nov} = \mathrm{Var} \otimes \mathrm{Nov}$ *then every* $D\mathrm{Var}$*-algebra is special.*

Proof. Suppose V is a $D\mathrm{Var}$-algebra. Then V may be presented as a quotient of a free algebra $D\mathrm{Var}\langle X\rangle$ modulo an ideal I. Consider the embedding $D\mathrm{Var}\langle X\rangle \subset \mathrm{Var}\langle X^{(\omega)}\rangle$ and denote by J the differential ideal of $\mathrm{Var}\langle X^{(\omega)}\rangle$ generated by I. Then $U = \mathrm{Var}\langle X^{(\omega)}\rangle/J$ is the universal enveloping differential Var-algebra of V. It remains to prove that $J \cap D\mathrm{Var}\langle X\rangle = I$, namely, the "$\subseteq$" part is not a trivial one.

Assume $f \in J$. Then there exists a family of (differential) polynomials $\Phi_i \in \mathrm{Var}\langle (X \cup \{t\})^{(\omega)}\rangle$ such that $f = \sum_i \Phi_i|_{t=g_i}$, for some $g_i \in I$. If, in addition, $f \in D\mathrm{Var}\langle X\rangle$ then $\mathrm{wt}(f) = -1$. Since $\mathrm{wt}\, g_i = -1$, we should have $\mathrm{wt}\, \Phi_i = -1$ for all i. By Lemma internal, every polynomial Φ_i may be represented as an element of $D\mathrm{Var}\langle X \cup \{t\}\rangle$, so $\Phi_i|_{t=g_i} \in I$ for all i, and thus $f \in I$. □

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